



Damage in rheumatic diseases: Contemporary international standpoint and scores emerging from clinical, radiological and machine learning

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ABSTRACT

In rheumatic diseases, damage is a major concern and reflects irreversible organ scarring or tissue degradation. Quantifying damage or measuring its severity is an indispensable concern in determining the overall outcome. Damage considerably influences both longterm prognosis and quality of life. Rheumatic diseases (RD) represent a significant health burden. Organ damage is consistently associated with increased mortality. Monitoring damage is critical in the evaluation of patients and in appraising treatment efficacy. Proper assessment and early detection of damage paves way for modifying the disease course with effective medications and regimens may reduce organ damage, improve outcomes and decrease mortality. With the exception of systemic lupus erythematosus and vasculitis, most RDs lack an established damage index making it an ongoing demand to develop effective scores and prediction models for damage accrual early in the disease course. A better understanding of machine learning with the increasing availability of medical large data may facilitate the development of meaningful precision medicine for patients with RDs. An updated spectrum of clinical and radiological damage scores and indices as well as the role of machine learning are presented in this review for the key RDs.

1. Introduction

The diagnosis of autoimmune rheumatic diseases (RDs) is usually paralleled by major concerns regarding the extent of organ damage caused by the inflammatory disease process. In rheumatology, the term damage is used to designate irreversible organ scarring or tissue degradation. Damage considerably influences the long-term disease prognosis, response to therapy and composes a significant burden on the quality of life of the affected patients. Quantifying damage or measuring

its severity is an essential concern in gauging the overall outcome with special reference to disease-related mortality [1]. RDs represent a significant health burden may be more than any medical subspecialty that affect a wide range of organ systems. They encompass a broad array of diseases including rheumatoid arthritis (RA), systemic lupus erythematosus (SLE), idiopathic inflammatory myositis (IIM), systemic sclerosis (SSc), vasculitis, Behçets disease and Sjögren's syndrome (SS) with a myriad spectrum of systemic manifestations that mandates the continuous quantification of disease severity and organ damage.

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Additionally, with the evolution of novel therapeutic modalities it seems crucial to continuously revise and appraise activity scores and damage indices [2]. A better understanding of machine learning with the increasing availability of medical large data may facilitate and support the development of meaningful precision medicine for patients with RDs.

2. Damage indices in autoimmune rheumatic diseases

2.1. Rheumatoid arthritis (RA)

Rheumatoid arthritis (RA) is an immune mediated inflammatory disease that primarily affects diarthrodial joints causing joint destruction and deformity. Therefore, in RA, the evaluation of damage mainly focuses on the extent of joint destruction which is considered the main cause of disability and incapacitation. Since the last century, joint damage is a major problem in the long-term course of RA and was usually assessed by plain radiography which was considered the gold standard for years [3]. Early radiological progression remains associated with long-term joint damage in real-world RA patients treated to the target of remission [4].

Plain x-ray: Most damage scores focus on the radiological structural changes. The findings related to disability in RA patients and indicating radiographic damage include true joint space narrowing (JSN), subluxation and erosions [5].

The *simplified erosion narrowing score* (SENS) has been found useful in clinical practice for estimating joint damage in early RA [6]. The rate of radiographic structural damage progression (*Ratingen erosion score*) was found helpful in determining clinical remission in patients taking biologics compared to synthetic disease modifying antirheumatic drugs (DMARDs) [7].

Long time ago, two methods of scoring radiographs have become standard techniques *Sharp* and *Larsen* indices and both concentrate on cartilage loss and erosive damage in the hands and wrists [4]. The *simplified Larsen erosion score* (SLES) has been developed for estimating joint damage in early RA in clinical practice [8].

Sharp score refers to an early radiographic scoring of erosions and JSN of hands and feet in RA first described in 1971 with a modified version proposed in 1985 by Sharp et al. [9] and was further improved by *van der Heijde*, also known as the *modified Total Sharp Score* (mTSS) and later *van der Heijde modification of the Sharp* scoring system (SvdH) [10]. The original Larsen score has been modified too [11]. Interestingly tacrolimus was found to have the potential to inhibit progression of joint damage in established RA as assessed by the mTSS [12]. Sensitivity to change of 5 scoring methods for radiographic damage on hand radiographs in patients with RA were compared including Larsen, Larsen/Rau, Sharp, SvdH and SENS. Sensitivity to change was higher with Sharp and SvdH for erosions and was less for JSN. The smallest detectable difference varied between 3.5 % (SvdH) and 14.2 % (SENS erosion) of their maximum scores [13].

The *Norfolk Arthritis Register Damaged Joint Count* (NOAR-DJC) in patients with early inflammatory polyarthritis is a quick, reliable and valid tool for assessing articular damage in relation to the eroded joint count [14]. The *digital x-ray radiogrammetry* (DXR) measurement of joint damage was found to be more precise than Sharp scores. DXR is simple to use, does not require intensive training, and may identify subjects not responding to standard treatment [15].

Musculoskeletal ultrasound (MSUS): MSUS assessment is a sensitive, reliable and feasible tool to detect cartilage damage [16] in RA, yet there is still emerging need for developing an ultrasound score to assess structural joint damage [17]. Interestingly, the severity of ultrasound-detected bone erosions was significantly greater when both positive power Doppler (PD) and a greater degree of grey scale (GS) joint inflammation were present. Single time point ultrasound joint inflammation assessment is reflective of joint damage severity in RA patients [18]. Higher PD grades penetrating bone are associated with

radiographic progression even in patients in remission [19]. Yet developing an ultrasound score for assessing structural joint damage in patients with RA is demanding.

Magnetic resonance imaging (MRI): The RA MRI score (RAMRIS) is a damage score used in clinical RA and combines joint erosions and JSN [20]. Furthermore, computer-assisted measurement of erosion volume had a good performance metrics and was more sensitive to change than RAMRIS [21]. The MRI erosion score [22] is valuable for assessing inflammatory and structural damage in patients with established RA receiving rituximab [23]. Structural damage at joints and entheses was commonly identified by whole body MRI and more frequently than by clinical examination. The MRI joint count is a promising tool for evaluation of the inflammatory load and structural damage in RA [24].

MRI and ultrasound of MCP and MTP joints are superior to x-ray to detect erosive disease in mild and moderate RA. MRI is slightly, but significantly more sensitive than ultrasound for erosive disease, while ultrasound is more sensitive to detect tenosynovitis. Both are comparably sensitive to detect synovitis [25].

Dual energy x-ray absorptiometry (DEXA): Bone damage was reported to be associated with bone loss in Moroccan RA patients [26].

Rheumatoid arthritis is not merely a disease of joints and typical extra-articular features include interstitial lung disease (ILD) and other co-morbidities such as osteoporosis, cardiovascular risk, recurrent infections and substantial cognitive dysfunction. Such complications may follow therapies or comorbidities [27]. Having an efficient damage index is crucial as a useful tool in clinical practice and research. Such improved indices should enhance care in management strategies. Moreover, certain subgroups of patients are under or over treated on existing therapeutic treat-to-target principles as co-morbidities and complications that lead to damage are not included in treatment algorithms. Redefining the management protocols would improve outcome and minimize damage [28].

It is about time to consider the development of well-structured indices that more widely reflect the damage accrual in RA. The *overall status in RA* (OSRA) involves 4 components one of them being disease damage score [29]. The *RA articular damage* (RAAD) score is another clinical method to assess long-term articular damage [30]. However, none of these scores has been designed to capture permanent global damage in RA. Damage in RA may be due to autoimmune and chronic inflammatory mechanisms, rising rates of comorbidities and treatment side effects. A call for a clinical index that involves different domains apart from joint damage is required to best reflect the global impact of the disease and could allow rheumatologists to spot the damage, stratify patients according to its grade, identify those with more aggressive disease and a poorer prognosis. Creating such an index for RA would allow personalized 'tailored' treatment with improved patient care [27].

Only slow-acting drugs such as gold have been persistently considered to reduce radiological progression and there was an ongoing call that rheumatologists look towards novel therapeutic approaches for a substantial reduction in the rate of damage [4]. Lately in RA there has been less articular damage and an improved long-term prognosis largely driven by enhanced treat-to-target and selective immune-targeted strategies, yet the frequency of challenging difficult-to-treat cases is not uncommon with the unavoidable consequences of damage [27]. Captivatingly, while the treatment paradigm of RA has changed over the years, based on several national and international sets of recommendations, there is an absence of a claim of inhibiting radiographic progression and structural damage in the product monograph of biologics [31].

In RA, possible biomarkers should help the rheumatologist to identify the patients who will develop structural damage more rapidly [32]. It is an ongoing dream to search for potentially promising biomarkers that would best reflect the structural joint damage and can predict longitudinal progression of damage.

2.2. Systemic lupus erythematosus (SLE)

Irreversible damage to organ systems occurs in SLE [33]. Indices that evaluate SLE damage include two patient reported forms; Lupus Damage Index Questionnaire (LDIQ) [34] and Brief Index of Lupus Damage (BILD) [35] based on the Systemic Lupus International Collaborating Clinics (SLICC) damage index (SDI) [36] that has been widely established and used. LDIQ versions in Spanish, Portuguese and French have been validated [37]. A revised SDI is under construction as shifts in the concept of damage in SLE have occurred with new insights into disease manifestations, diagnostics, and therapy [38]. A modified version of the SDI for use in children and adolescents with SLE has been devised, based on the frequency and distribution of damage in patients with juvenile-onset SLE and the sources of damage that are most suitable for inclusion in a pediatric damage index [39]. Organ damage accrual was defined as a change of SDI score > 0 units between baseline and final annual visits [40]. Knowing that damage will most likely happen after the first 6 years from the diagnosis in 50 % of patients enables physicians to better predict damage occurrence [41]. Table 1 presents the SDI differences in SLE and juvenile SLE (JSLE) patients across various countries [42–59]. Interestingly, the SDI was comparable in JSLE while it was relatively higher in adults from Egypt, which may reflect the impact of associated comorbidities that evolve over time.

The recently developed SLICC frailty index (SLICC-FI) was found to be associated with damage accrual (SDI) in SLE patients supporting the importance of this index in the evaluation of SLE patients [60].

In addition to its role in disease classification, the EULAR/ACR SLE criteria may have value in predicting the prognosis and damage as the renal and gonadal domains were strongly related to the SDI [61]. The high number of the ACR/EULAR criteria and treatment with steroids in childhood-onset SLE are associated with damage accrual and these findings could enable us to detect patients who should be closely monitored for higher risk of damage development [41]. Low disease activity and remission, on and off-treatment, are associated with less damage accrual [62].

Organ damage in SLE (increased SDI) is consistently associated with increased mortality across studies from various countries. Modifying the disease course with effective therapies and steroid-sparing regimens may reduce organ damage, improve outcomes and decrease mortality

for patients with SLE [63].

Antiphospholipid syndrome (APS): A damage index specific for APS (DIAPS) was developed [64]. A distinct pattern of damage is present in primary APS (PAPS) and APS related to SLE. Damage in PAPS is an early event, while APS-SLE is associated with higher long-term damage, with a striking increment of damage over time. A diagnosis delay is related to higher damage. Damage surveillance therefore requires different approaches for these two conditions [65].

2.3. Systemic sclerosis (SSc)

In systemic sclerosis (SSc), damage is mainly contributed to the multisystem fibrosing nature of the disease with microvascular involvement affecting the skin and internal organs, damage in SSc is defined as irreversible/persistent changes (>6 months) due to previous active disease/complications of therapy [66]. Monitoring damage is critical in the evaluation of patients and in appraising treatment efficacy. A disease damage (severity) scale for SSc is required [67]. Damage accrual in SSc can be tracked using the Scleroderma Clinical Trials Consortium Damage Index [68]. Yet it remains important to early differentiate disease activity from damage in SSc [69].

Localized scleroderma (LoS) is a chronic fibrosing disorder of the skin and, less commonly, subcutaneous tissues. As the disease causes subjective symptoms, cosmetic defects, and, at times, functional disability, subjects with LoS experience deterioration of their health-related quality of life (HRQoL). The damage status in adult patients with LoS could be assessed using physician-reported measures as the LoS skin damage index, physician global assessments of the disease damage and patient-derived measures as the patient global assessments of the disease damage [70]. *LoS Skin Damage Index (LoSDI)* was early developed and validated. LoSDI was calculated by summing three scores for cutaneous features of damage: dermal atrophy (DAT), subcutaneous atrophy (SAT) and dyspigmentation (DP) measured at 18 anatomic sites. LoSDI is moderately associated with *Physician Global Assessment of disease Damage* (PGA-D) and poorly with Patient GA of disease severity (PtGA-S) [66]. However, the need for more reliable and effective scores goes on.

Microvascular damage is assessed by nailfold videocapillaroscopy and skin impairment evaluated by three different methods; the *modified*

Table 1
Comparison of the systemic lupus international collaborating clinics (SLICC) damage index (SDI) in adult and juvenile systemic lupus erythematosus patients in different countries.

Country/reference	Year	SDI score	n	Age (y)	Age at onset (y)	Gender F:M
SLE						
Italy [42]	2022	1.57 ± 1.9	221	28.5 ± 6.6	25.6 ± 10.6	8.6:1
Spain [43]	2020	1.1 ± 1.6	2398	46.1 ± 14.2	34.2 ± 11.4	9.4:1
Sweden [44]	2020	1.4	340	46.7	29.1	7.4:1
USA [45]	2015	1.2	2199	38	32.9	12.3:1
Canada [46]	2021	2 ± 1.5	649	37.6 ± 13.6	30.3 ± 12.9	9.8:1
Mexico [47]	2023	1.1	200	44.9 ± 13.2	31.2 ± 12	12.3:1
Peru [48]	2022	1.7 ± 1.7	265	–	35.1 ± 13.6	14.6:1
China [49]	2018	0.33 ± 0.66	577	41.2 ± 13.4	31.9 ± 10.3	13.8:1
Korea [50]	2021	0.99 ± 1.1	259	34 ± 13.7	25.4 ± 9.1	12:1
Japan [51]	2023	0 (0–1)	702	46 (35–58)	38.7 (31–47)	6.4:1
Malaysia [52]	2016	≥1 (43 %)	424	38.9 ± 13.5	29.7 ± 13.2	11.5:1
Turkey [53]	2017	2.2 ± 1.6	719	–	34 ± 11.3	6.1:1
Egypt [54]	2021	2.3 ± 2.5	3661	32.4 ± 10.1	26.5 ± 9.2	8.8:1
Australia [55]	2023	1 (0–7)	89	45 (22–64)	29.7 (22–25.3)	11.7:1
JSLE						
UK [56]	2021	0.6	428	–	12.2 (0.1–17.9)	5.4:1
Thailand [57]	2023	≥1 (40 %)	232	–	12.2	6:1
Oman [58]	2019	0.76 ± 1.38	103	–	10.4	4:1
Turkey [53]	2017	1.9 ± 1.2	216	–	13.7 ± 3.5	6.7:1
Egypt [59]	2021	0.36 ± 0.78	404	13.2 ± 2.4	11.1 ± 2.5	7.2:1

SDI: Systemic Lupus International Collaborating Clinics (SLICC) damage index, SLE: systemic lupus erythematosus, JSLE: juvenile SLE.

Rodnan skin score (mRSS), skin high-frequency ultrasound (US) and the plicometer skin test (PST) in SSc patients [71]. It reflects well the histological grading of skin fibrosis however; it is unable to identify small changes. Although RA-like hand radiological damage can be observed in SSc patients, arthropathies are less common than previously reported [72]. More recently, efforts were done to develop the first damage index in SSc through the development and validation of a 23-item *Scleroderma Clinical Trials Consortium Damage Index* (SCTC-DI) [68].

2.4. Idiopathic inflammatory myositis (IIM)

Damage in IIM is primarily attributed to muscle inflammation with an associated microvascular process amongst other systemic manifestations. There are few outcome measures in myositis that are not validated and assessment of outcomes in therapeutic trials was focused on non-standardized measurement of muscle strength and function only [73].

Physician global damage is an overall rating of the disease damage related to myositis and is defined as persistent changes in anatomy, pathology, physiology, or function, such as fibrosis, scarring, or atrophy, resulting from any cause (including prior treatment) since the onset of the myositis and present for at least 6 months despite immunosuppressive or other therapy, including exercise and rehabilitation. It has been included in the assessment of damage for adult and juvenile patients with polymyositis (PM), dermatomyositis (DM), and inclusion body myositis (IBM) by the International Myositis Assessment and Clinical Studies Group (IMACS) [74].

The *myositis damage index* (MDI) has been devised to assess the extent and severity of damage developing in different organs and systems should enhance the consistency, comprehensiveness and reliability of damage assessment in patients with myositis. It is valuable in assessing damage in prospective clinical trials in adult IIM patients [75].

2.5. Spondyloarthritis (SpA)

Whole body MRI was suggested to allow simultaneous assessment of peripheral and axial joints in SpA, to present the distribution of inflammatory and structural lesions and to determine global scores, however there has always been a need for specific effective damage scores that may potentially serve for follow up [76].

Early assessment of bone marrow oedema on MRI of the sacroiliac joints (MRI-SIJ) is important to alarm before the development of structural damage on radiographs of the SIJ (X-SIJ) in axial SpA (axSpA) [77].

The performance of different spinal radiographic damage scoring methods were compared in patients with early axSpA; the *DEvenir des Spondylarthropathies Indifférenciées Récentes* (DESIR) assessed *Stoke AS Spine Score* (SASSS) [78], *modified SASSS* (mSASSS), *radiographic ankylosing spondylitis (AS) spinal score* (RASSS) [79], *Bath AS radiographic index* (BASRI-spine and -total) [80] and existing scores assessing spinal radiographic damage performed well in early phases of axSpA. Chronic inflammation, structural damage affecting the spine and sacroiliac joints and treatment response in AxSpA may be assessed using the established Spondyloarthritis Research Consortium of Canada (SPARCC) inflammation and sacroiliac joint structural (SSS) scoring methods, which are two MRI-based scoring methods [81].

In axSpA, SIJ erosion is often followed by fat metaplasia in an erosion cavity (backfill), and subsequently ankylosis. The potential clinical implication is assessment of SIJ damage progression by 1 composite score. In a newer attempt, SPARCC SIJ structural score for erosion, backfill, and ankylosis were combined into 3 versions of a novel preliminary axSpA MRI SIJ *Composite Structural Damage Score* (CSDS). CSDS showed construct validity and responsiveness [82].

2.6. Vasculitis

The concept of damage is an important outcome measure for clinical trials. Because most vasculitis flares can be treated effectively in the short term, achievement of remission by itself may not be an adequate endpoint. Damage indicates the cost of achieving remission, aggregating the chronic morbidity that occurs after the disease flare has been treated. If 2 drugs both induce remission within 6 months of initiating therapy, the drug that also prevents the development of damage is superior. Damage thus represents a unique domain that is not captured by other domains of vasculitis, including disease activity, quality of life, and other patient-reported outcomes [83].

Both *vasculitis damage index* (VDI) [84] and the *combined damage assessment index* (CDA) [85] measure damage from vasculitis. The CDA captures more detail but is more complex and less practical than VDI. An *ANCA-associated vasculitis index of damage* (AVID) records damage unique to granulomatosis with polyangiitis (GPA) and microscopic polyangiitis (MPA) [86]. Further evolution of damage assessment in vasculitis to include key elements is warranted. A damage assessment tool, *Combined Arteritis Damage Score* (CARDS) [87] based on imaging in large-vessel vasculitis (LVV) of potential value to clinical studies and management has been developed. Takayasu (TAK) and giant cell arteritis (GCA) differ in the radiologic severity of disease.

Measures of damage in vasculitis do not account for the possibility that some forms of damage may exert greater impact than others. Revision by experts on how damage is quantified in vasculitis clinical research is demanded [88]. Damage denotes the aspects of chronic disease that do not reverse with therapy. This concept is important for primary systemic vasculitides, since the careful differentiation between activity and damage may help avoid unnecessary exposure to cytotoxic medications. Damage significantly influences both long-term prognosis and quality of life. Because primary systemic vasculitides have diverse manifestations, the use of a damage assessment tool is crucial to ensure reproducibility. VDI is the only validated measure for damage assessment in vasculitis but may not adequately determine the full spectrum of damage experienced by patients with vasculitis of small- and medium-size vessels [85]. Current therapy has transformed the prognosis of systemic vasculitides from diseases that had a high acute mortality rate to chronic relapsing diseases with high rates of morbidity. This improved survival highlighted the need for better methods of monitoring and recording the accumulation of organ damage during the disease course [89].

2.7. Behçet's disease (BD)

Recently the Behçet's Syndrome Overall Damage Index (BODI) was developed with pending validation [90] while the BD damage index (BDI) was concordantly developed and validated [91]. The BDI was found to be 5.1 ± 2.9 in a large cohort of 1252 BD patients with a mean age of 36.1 ± 9.9 and M: F 2.6:1 extracted from a nationwide study from Egypt [92].

The BD damage indices, VDI, BODI and BDI, have good convergent and discriminative validity for the assessment of BD damage and the three indices are highly correlated; however, BDI showed more sensitivity than BODI in detecting the damage accrual and showed more agreement with VDI [93].

3. Role of machine learning (ML)

A better understanding of ML with the increasing availability of medical large data may facilitate the development of meaningful precision medicine for patients with RDs [94]. A simple prediction model for joint damage progression in early RA has been proposed by only counting the number of risk factors and was found to have a satisfactory performance compared to an extended model [95].

On applying recurrent neural networks (RNNs) as a ML model to

predict the risk of chronic damage in SLE, patients at potential risk were identified [96]. In Spain, on a large cohort of SLE patients, cardiovascular and musculoskeletal damage manifestations were the two dominant forms of damage to sort patients into clinically meaningful clusters. Both in early and late stages of the disease, there was a significant association of these clusters with an increased risk of mortality [97]. The SDI is associated with increased healthcare costs and mortality and comparing the trajectory of individual damage items according to ethnicities in SLE patients has proved effective [98]. Recently, damage clusters evidence has been obtained for subphenotypes of SLE [33]. Autoantibody clustering is a valuable tool to differentiate various subsets of SLE, allowing prediction of subsequent clinical course and organ damage. Using SDI, compared to patients with cluster 1 (anti-Sm and anti-RNP), the greatest frequency of nephrotic syndrome was observed in patients in cluster 2 (anti-dsDNA, anti-Ro, and anti-La), whereas cluster 3 patients (anti-dsDNA, LAC, and aCL) had highest percentage of damage due to cerebrovascular accident and venous thrombosis. Osteoporotic fracture was also more common in cluster 3 than in cluster 2 [99].

To develop a prediction model for damage accrual in SSc patients with early disease is a chief goal. A simple model using only cutaneous subset, baseline damage index and sex can predict damage accumulation in early SSc [68].

Eye lesions, occur in nearly half of patients with BD, can lead to irreversible damage and vision loss. Using information obtained in the clinical settings, extreme gradient boosting identified patients at higher risk of vision-threatening BD (VTBD) [100].

In conclusion, quantifying damage or measuring its severity is an indispensable concern in determining the overall outcome in rheumatic diseases. Monitoring damage is critical in the evaluation of patients and in appraising treatment efficacy. Proper assessment and early detection of damage paves way for modifying the disease course and effective therapies and regimens may reduce organ damage, improve outcomes and decrease mortality.

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References

- [1] Holland MJ, Beresford MW, Feldman BM, Huggins J, Norambuena X, Silva CA, et al. Measuring disease damage and its severity in childhood-onset systemic lupus erythematosus. *Arthritis Care Res (Hoboken)* 2018;70(11):1621–9.
- [2] Kono DH, Theofilopoulos AN. Autoimmunity and tolerance. In: Firestein & Kelley's Textbook of Rheumatology. Firestein GS, Budd RC, Gabriel SE, Koretzky GA, McInnes IB, O'Dell JR. (eds). Elsevier, Philadelphia, 11th edition. 2021; Chapter 22, 337–54.
- [3] Scott DL, Bacon PA. Joint damage in rheumatoid arthritis: radiological assessments and the effects of anti-rheumatic drugs. *Rheumatol Int* 1985;5(5): 193–9.
- [4] Versteeg GA, Steunebrink L, Vonkeman HE, Ten Klooster PM, Van Der Bijl AE, Van De Laar M. Early radiological progression remains associated with long-term joint damage in real-world rheumatoid arthritis patients treated to the target of remission. *Scand J Rheumatol* 2022;51(2):87–96.
- [5] Navarro-Compán V, Landewé R, Provan SA, Ødegård S, Uhlig T, Kvien TK, et al. Relationship between types of radiographic damage and disability in patients with rheumatoid arthritis in the EURIDISS cohort: a longitudinal study. *Rheumatology (Oxford)* 2015;54(1):83–90.
- [6] Forslind K; BARFOT Study Group. The usefulness of the simplified erosion narrowing score (SENS) in clinical practice for estimating joint damage in early rheumatoid arthritis. *Scand J Rheumatol* 2011;40(6):497–8.
- [7] Ciubotariu E, Gabay C, Finckh A. Physicians of the Swiss clinical quality management program for rheumatoid arthritis. Joint damage progression in patients with rheumatoid arthritis in clinical remission: do biologics perform better than synthetic antirheumatic drugs? *J Rheumatol* 2014;41(8):1576–82.
- [8] Ziegeler M, Forslind K. Simplified Larsen erosion score (SLES) for estimating joint damage in early rheumatoid arthritis in clinical practice. *Scand J Rheumatol* 2021;50(5):402–3.
- [9] Sharp JT, Young DY, Bluhm GB, Brook A, Brower AC, Corbett M, et al. How many joints in the hands and wrists should be included in a score of radiologic abnormalities used to assess rheumatoid arthritis? *Arthritis Rheum* 1985;28(12): 1326–35.
- [10] Van der Heijde D. How to read radiographs according to the Sharp/van der Heijde method. *J Rheumatol* 1999;26:743–517.
- [11] Rau R, Herborn G. A modified version of Larsen's scoring method to assess radiologic changes in rheumatoid arthritis. *J Rheumatol* 1995;22:1976–82.
- [12] Motomura H, Matsushita I, Seki E, Mine H, Kimura T. Inhibitory effect of tacrolimus on progression of joint damage in patients with rheumatoid arthritis. *Int J Rheum Dis* 2014;17(7):749–54.
- [13] Guillemin F, Billot L, Boini S, Gerard N, Ødegård S, Kvien TK. Reproducibility and sensitivity to change of 5 methods for scoring hand radiographic damage in patients with rheumatoid arthritis. *J Rheumatol* 2005;32(5):778–86.
- [14] Bunn DK, Shepstone L, Galpin LM, Wiles NJ, Symmons DP. The NOAR Damaged Joint Count (NOAR-DJC): a clinical measure for assessing articular damage in patients with early inflammatory polyarthritis including rheumatoid arthritis. *Rheumatology (Oxford)* 2004;43(12):1519–25.
- [15] Jawaid WB, Crosbie D, Shotton J, Reid DM, Stewart A. Use of digital x ray radiogrammetry in the assessment of joint damage in rheumatoid arthritis. *Ann Rheum Dis* 2006;65(4):459–64.
- [16] Cipolletta E, Mandl P, Di Matteo A, Mashadi Mirza R, Passarini G, Grassi W, et al. Sonographic assessment of cartilage damage at the metacarpal head in rheumatoid arthritis: qualitative versus quantitative methods. *Rheumatology (Oxford)* 2022;61(3):1018–25.
- [17] Gessl I, Balint PV, Filippucci E, Keen HI, Pineda C, Terslev L, et al. Structural damage in rheumatoid arthritis assessed by musculoskeletal ultrasound: A systematic literature review by the Structural Joint Damage Task Force of the OMERACT Ultrasound Working Group. *Semin Arthritis Rheum* 2021;51(3): 627–39.
- [18] Tan YK, Li H, Allen Jr JC, Thumboo J. Joint damage in rheumatoid arthritis: Lessons learned from an extended 36-joint ultrasonography and 28-joint disease activity score assessments. *Int J Rheum Dis* 2021;24(1):106–11.
- [19] Raffaeiner B, Grisan E, Botsios C, Stramare R, Rizzo G, Bernardi L, et al. Grade and location of power Doppler are predictive of damage progression in rheumatoid arthritis patients in clinical remission by anti-tumour necrosis factor α . *Rheumatology (Oxford)* 2017;56(8):1320–5.
- [20] Sundin U, Østergaard M, Glinatsi D, Aga AB, Horslev-Petersen K, Hetland ML, et al. Validity and responsiveness of combined inflammation and combined joint damage scores based on the OMERACT rheumatoid arthritis MRI scoring system (RAMRIS). *J Rheumatol* 2019;46(9):1222–7.
- [21] Poh MQ, Lassere M, Bird P, Edmonds J. Reliability and longitudinal validity of computer-assisted methods for measuring joint damage progression in subjects with rheumatoid arthritis. *J Rheumatol* 2013;40(1):23–9.
- [22] Østergaard M, Peterfy C, McQueen CP, Bird F, Ejbjerg PB, et al. OMERACT rheumatoid arthritis magnetic resonance imaging studies. Core set of MRI acquisitions, joint pathology definitions and the OMERACT RA-MRI scoring system. *J Rheumatol* 2003;30:1385–6.
- [23] Peterfy C, Emery P, Tak PP, Østergaard M, DiCarlo J, Otsa K, et al. MRI assessment of suppression of structural damage in patients with rheumatoid arthritis receiving rituximab: results from the randomised, placebo-controlled, double-blind RA-SCORE study. *Ann Rheum Dis* 2016;75(1):170–7.
- [24] Axelsen MB, Eshed I, Duer-Jensen A, Møller JM, Pedersen SJ, Østergaard M. Whole-body MRI assessment of disease activity and structural damage in rheumatoid arthritis: first step towards an MRI joint count. *Rheumatology (Oxford)* 2014;53(5):845–53.
- [25] Schmidt WA, Schicke B, Ostendorf B, Scherer A, Krause A, Walther M. Low-field MRI versus ultrasound: which is more sensitive in detecting inflammation and bone damage in MCP and MTP joints in mild or moderate rheumatoid arthritis? *Clin Exp Rheumatol* 2013;31(1):91–6.
- [26] Azzouzi H, Ichchou L. Bone loss and radiographic damage profile in rheumatoid arthritis Moroccan patients. *J Bone Metab* 2021;28(2):151–9.
- [27] Fragoulis GE, Chatziodionysiou K, Nikiforou E, Cope A, McInnes IB. Damage accrual in rheumatoid arthritis: evaluating the joint and beyond. *Arthritis Rheumatol* 2020;72(12):1967–70.
- [28] Landewé RBM. Overdiagnosis and overtreatment in rheumatology: a little caution is in order. *Ann Rheum Dis* 2018;77(10):1394–6.
- [29] Symmons DP, Hassell AB, Gunatillaka KA, Jones PJ, Schollum J, Dawes PT. Development and preliminary assessment of a simple measure of overall status in rheumatoid arthritis (OSRA) for routine clinical use. *QJM* 1995;88(6):429–37.
- [30] Zijlstra TR, Bernelot Moens HJ, Bukhari MA. The rheumatoid arthritis articular damage score: first steps in developing a clinical index of long term damage in RA. *Ann Rheum Dis* 2002;61(1):20–3.
- [31] Haraoui B, Karsh J, Pope JE, Thorne JC, Keystone EC. Is structural damage evaluation by traditional radiographs still relevant in rheumatoid arthritis clinical trials? *J Rheumatol* 2014;41(11):2325.
- [32] Ferraccioli G, Alivernini S, Gremese E. Biomarkers of joint damage in rheumatoid arthritis: where are we in 2013? *J Rheumatol* 2013;40(8):1244–6.
- [33] Ahn GY, Lee J, Won S, Ha E, Kim H, Nam B, et al. Identifying damage clusters in patients with systemic lupus erythematosus. *Int J Rheum Dis* 2020;23(1):84–91.

- [34] Costenbader KH, Khamashta M, Ruiz-Garcia S. Multicenter validation of the self-reported lupus damage index questionnaire (LDIQ). *Arthritis Rheum* 2008;58: S802–3.
- [35] Yazdany J, Trupin L, Gansky SA, Dall'era M, Yelin EH, Criswell LA, et al. Brief index of lupus damage: a patient-reported measure of damage in systemic lupus erythematosus. *Arthritis Care Res (Hoboken)* 2011;63:1170–7.
- [36] Gladman D, Ginzler E, Goldsmith C, Fortin P, Liang M, Urowitz M, et al. The development and initial validation of the Systemic Lupus International Collaborating Clinics/American College of Rheumatology damage index for systemic lupus erythematosus. *Arthritis Rheum* 1996;39:363–9.
- [37] Pons-Estel BA, Sanchez-Guerrero J, Romero-Diaz J, Iglesias-Gamarra A, Bonfa E, Borba EF, et al. Validation of the Spanish, Portuguese and French versions of the lupus damage index questionnaire: data from North and South America, Spain and Portugal. *Lupus* 2009;18:1033–52.
- [38] Johnson SR, Gladman DD, Brunner HI, Isenberg D, Clarke AE, Barber MR, et al. Evaluating the construct of damage in systemic lupus erythematosus. *Arthritis Care Res (Hoboken)*. 2021; epub ahead of print.
- [39] Gutiérrez-Suárez R, Ruperto R, Gastaldi R, Pistorio A, Felici E, Burgos-Vargas R, et al. A proposal for a pediatric version of the systemic lupus international collaborating clinics/American college of rheumatology damage index based on the analysis of 1,015 patients with juvenile-onset systemic lupus erythematosus. *Arthritis Rheum* 2006;54(9):2989–96.
- [40] Kandane-Rathnayake R, Kent JR, Louthrenoo W, Luo SF, Wu YJ, Lateef A, et al. Longitudinal associations of active renal disease with irreversible organ damage accrual in systemic lupus erythematosus. *Lupus* 2019;28(14):1669–77.
- [41] Kifer N, Sestan M, Frkovic M, Kifer D, Kozmar A, Padjen I, et al. 2019 ACR/EULAR classification criteria and therapy in predicting organ damage accrual in patients with childhood-onset systemic lupus erythematosus: A retrospective study over the last 29 years. *Lupus* 2022;31(7):828–36.
- [42] Gerosa M, Beretta L, Ramirez GA, Bozzolo E, Cornalba M, Bellocchi C, et al. Long-term clinical outcome in systemic lupus erythematosus patients followed for more than 20 years: The Milan systemic lupus erythematosus consortium (SMiLE) Cohort. *J Clin Med* 2022;11(13):3587.
- [43] Riancho-Zarrabeitia L, Martínez-Taboada V, Rúa-Figueroa I, Alonso F, Galindo-Izquierdo M, Ovalles J, et al. Antiphospholipid syndrome (APS) in patients with systemic lupus erythematosus (SLE) implies a more severe disease with more damage accrual and higher mortality. *Lupus* 2020;29(12):1556–65.
- [44] Elbagir S, Elshafie AI, Elagib EM, Mohammed NA, Aledrissy MIE, Sohrabian A, et al. Sudanese and Swedish patients with systemic lupus erythematosus: immunological and clinical comparisons. *Rheumatology (Oxford)* 2020;59(5): 968–78.
- [45] Al Sawah S, Zhang X, Zhu B, Magder LS, Foster SA, Iikuni N, et al. Effect of corticosteroid use by dose on the risk of developing organ damage over time in systemic lupus erythematosus—the Hopkins Lupus Cohort. *Lupus Sci Med* 2015;2 (1):e000066.
- [46] Urowitz MB, Gladman DD, Ibañez D, Su J, Mursleen S, Sayani A, et al. Effect of disease activity on organ damage progression in systemic lupus erythematosus: University of Toronto lupus clinic cohort. *J Rheumatol* 2021;48(1):67–73.
- [47] López-Villanueva RF, Valencia-Pacheco G, Zapata-Vázquez R, López-Suárez R, Castro-Sansores C. Follow-up of clinical activity and accumulated organ damage in a cohort of patients with systemic lupus erythematosus from the Yucatán Peninsula, Mexico (1995–2016). *Rheumatol Clin (Engl Ed)* 2023;19(2):106–13.
- [48] Elera-Fitzcarrald C, Gamboa-Cárdenas RV, Reategui-Sokolova C, Pimentel-Quiroz V, Rodriguez-Bellido Z, Pastor-Asurza CA, et al. The SLICC-FI predicts damage accrual in SLE patients. Data from the Almenara lupus cohort. *Lupus* 2022;31(13):1666–70.
- [49] Mok CC, Tse SM, Chan KL, Ho LY. Effect of the metabolic syndrome on organ damage and mortality in patients with systemic lupus erythematosus: a longitudinal analysis. *Clin Exp Rheumatol* 2018;36(3):389–95.
- [50] Park DJ, Yoon CS, Choi SE, Xu H, Kang JH, Lee SS. Risk factors for thrombotic events in Korean patients with systemic lupus erythematosus. *Sci Rep* 2021;11(1): 23529.
- [51] Yanai R, Yajima N, Oguro N, Shimojima Y, Ohno S, Kajiyama H, et al. Number of attending physicians and accumulated organ damage in patients with systemic lupus erythematosus: LUNA Registry cross-sectional study. *Rheumatol Ther* 2023; 10(2):421–31.
- [52] Shaharir SS, Hussein H, Rajalingham S, Mohamed Said MS, Abdul Gafor AH, Mohd R, et al. Damage in the multiethnic Malaysian systemic lupus erythematosus (SLE) cohort: Comparison with other cohorts worldwide. *PLoS One* 2016;11(11):e0166270.
- [53] Artim-Esen B, Şahin S, Çene E, Şahinkaya Y, Barut K, Adrovic A, et al. Comparison of disease characteristics, organ damage, and survival in patients with juvenile-onset and adult-onset systemic lupus erythematosus in a combined cohort from 2 tertiary centers in Turkey. *J Rheumatol* 2017;44(5):619–25.
- [54] Gheita TA, Noor RA, Abualfadi E, Abousehly OS, El-Gazzar II. Egyptian college of rheumatology (ECR) SLE study group. Adult systemic lupus erythematosus in Egypt: The nation-wide spectrum of 3661 patients and world-wide standpoint. *Lupus* 2021;30(9):1526–35.
- [55] Raghunath S, Glikmann-Johnston Y, Golder V, Kandane-Rathnayake R, Morand EF, Stout JC, et al. Clinical associations of cognitive dysfunction in systemic lupus erythematosus. *Lupus Sci Med* 2023;10(1):e000835.
- [56] Giani T, Smith EM, Al-Abadi E, Armon K, Bailey K, UK JSLE Cohort Study. Neuropsychiatric involvement in juvenile-onset systemic lupus erythematosus: Data from the UK Juvenile-onset systemic lupus erythematosus cohort study. *Lupus* 2021;30(12):1955–65.
- [57] Na Nakorn K, Piyaphanee N, Sukharomana M, Pinpatanapong R, Charuvani J S. Outcomes of achieving lupus low disease activity state and damage accrual in childhood-onset systemic lupus erythematosus. *Clin Rheumatol* 2023. epub ahead of print.
- [58] Abdwani R, Abdalla E, Al-Zakwani I. Unique characteristics of prepubertal onset systemic lupus erythematosus. *Int J Pediatr* 2019;2019:9537065.
- [59] Eesa NN, Abdel Nabi H, Owaidey RE, Khalifa I, Radwan AR, Egyptian College of Rheumatology (ECR) SLE Study Group. Systemic lupus erythematosus children in Egypt: Homeland spectrum amid the global situation. *Lupus* 2021;30(13): 2135–43.
- [60] Ugarte-Gil MF, Dubey J, McGwin Jr G, Reveille JD, Vilá LM, Alarcón GS. Association of systemic lupus international collaborating clinics frailty index with damage in systemic lupus erythematosus patients: Results from a multiethnic, multicenter US cohort of patients with lupus. *Arthritis Care Res (Hoboken)* 2023; 75(3):585–9.
- [61] Mok CC, Chung YK, Lee C, Ho LY, To CH. Relationship between the EULAR/ACR classification criteria and organ damage in systemic lupus erythematosus. *Lupus* 2023;32(3):424–30.
- [62] Ugarte-Gil MF, Hanly J, Urowitz M, Gordon C, Bae SC, Romero-Diaz J, et al. Remission and low disease activity (LDA) prevent damage accrual in patients with systemic lupus erythematosus: results from the systemic lupus international collaborating clinics (SLICC) inception cohort. *Ann Rheum Dis* 2022;81(11): 1541–8.
- [63] Murimi-Worstell IB, Lin DH, Nab H, Kan HJ, Onasanya O, Tierce JC, et al. Association between organ damage and mortality in systemic lupus erythematosus: a systematic review and meta-analysis. *BMJ Open* 2020;10(5): e031850.
- [64] Amigo MC, Goycochea-Robles MV, Espinosa-Cuervo G, Medina G, Barragán-Garfias JA, Vargas A, et al. Development and initial validation of a damage index (DIAPS) in patients with thrombotic antiphospholipid syndrome (APS). *Lupus* 2015;24:927–34.
- [65] Torricelli AK, Ugolini-Lopes MR, Bonfá E, Andrade D. Antiphospholipid syndrome damage index (DIAPS): distinct long-term kinetic in primary antiphospholipid syndrome and antiphospholipid syndrome related to systemic lupus erythematosus. *Lupus* 2020;29(3):256–62.
- [66] Arkachaisri T, Vilaiyuk S, Torok KS, Medsger Jr TA. Development and initial validation of the localized scleroderma skin damage index and physician global assessment of disease damage: a proof-of-concept study. *Rheumatology (Oxford)* 2010;49(2):373–81.
- [67] Medsger Jr TA. Assessment of damage and activity in systemic sclerosis. *Curr Opin Rheumatol* 2000;12(6):545–8.
- [68] Ferdowsi N, Huq M, Stevens W, Hudson M, Wang M, Tay T, et al. Development and validation of the scleroderma clinical trials consortium damage index (SCTC-DI): a novel instrument to quantify organ damage in systemic sclerosis. *Ann Rheum Dis* 2019;78(6):807–16.
- [69] Jain S, Sharma SK. Differentiating disease activity from damage in systemic sclerosis: it's still early days! *Ann Rheum Dis* 2020;79:e98.
- [70] Lis-Święty A, Skrzypek-Salamon A, Ranoš-Janicka I, Brzezińska-Wcisło L. Associations between disease activity/severity and damage and health-related quality of life in adult patients with localized scleroderma-A Comparison of LoSCAT and visual analogue scales. *J Clin Med* 2020;9(3):756.
- [71] Ruaro B, Pizzorni C, Paolino S, Smith V, Ghio M, Casabella A, et al. Correlations between nailfold microvascular damage and skin involvement in systemic sclerosis patients. *Microvasc Res* 2019;125:103874.
- [72] Koutaissoff S, Vanthuyne M, Smith V, De Langhe E, Depressieux G, Westhovens R, et al. Hand radiological damage in systemic sclerosis: comparison with a control group and clinical and functional correlations. *Semin Arthritis Rheum* 2011;40 (5):455–60.
- [73] Rider LG, Werth VP, Huber AM, Alexanderson H, Rao AP, Ruperto N, et al. Measures of adult and juvenile dermatomyositis, polymyositis, and inclusion body myositis: Physician and patient/parent global activity, manual muscle testing (MMT), health assessment questionnaire (HAQ)/childhood health assessment questionnaire (C-HAQ), childhood myositis assessment scale (CMAS), myositis disease activity assessment tool (MDAAT), disease activity score (DAS), short form 36 (SF-36), child health questionnaire (CHQ), physician global damage, myositis damage index (MDI), quantitative muscle testing (QMT), myositis functional index-2 (FI-2), myositis activities profile (MAP), inclusion body myositis functional rating scale (IBMFRS), cutaneous dermatomyositis disease area and severity index (CDASI), cutaneous assessment tool (CAT), dermatomyositis skin severity index (DSSI), Skindex, and dermatology life quality index (DLQI). *Arthritis Care Res (Hoboken)*. 2011;63 Suppl 11(0 11):S118–57.
- [74] Miller FW, Rider LG, Chung YL, Cooper R, Danko K, Farewell V, et al. Proposed preliminary core set measures for disease outcome assessment in adult and juvenile idiopathic inflammatory myopathies. *Rheumatology (Oxford)* 2001;40: 1262–73.
- [75] Sultan SM, Allen E, Cooper RG, Agarwal S, Kiely P, Oddis CV, et al. Interrater reliability and aspects of validity of the myositis damage index. *Ann Rheum Dis* 2011;70(7):1272–6.
- [76] Poggenborg RP, Pedersen SJ, Eshed I, Sørensen IJ, Møller JM, Madsen OR, et al. Head-to-toe whole-body MRI in psoriatic arthritis, axial spondyloarthritis and healthy subjects: first steps towards global inflammation and damage scores of peripheral and axial joints. *Rheumatology (Oxford)* 2015;54(6):1039–49.
- [77] Sepriano A, Ramiro S, Landewé R, Dougados M, van der Heijde D, Rudwaleit M. Is active sacroiliitis on MRI associated with radiographic damage in axial spondyloarthritis? Real-life data from the ASAS and DESIR cohorts. *Rheumatology (Oxford)* 2019;58(5):798–802.

- [78] Aversn HL, Oxtoby J, Jones THG, PW, Dziedzic K, Dawes PT.. Radiological outcome in ankylosing spondylitis: use of the Stoke ankylosing spondylitis spine score (SASSS). *Br J Rheumatol* 1996;35:373–6.
- [79] Creemers MC, Franssen MJ, van't Hof MA, Gribnau FW, van de Putte LB, van Riel PL. Assessment of outcome in ankylosing spondylitis: an extended radiographic scoring system. *Ann Rheum Dis* 2005;64:127–9.
- [80] MacKay K, Brophy S, Mack C, Doran M, Calin A. The development and validation of a radiographic grading system for the hip in ankylosing spondylitis: the Bath ankylosing spondylitis radiology hip index. *J Rheumatol* 2000;27:2866–72.
- [81] Zhang Y, Guo Z, Zhan Y, Qu J, Lei X. Evaluation of active inflammation, chronic structural damage, and response to treatment of sacroiliitis in axial spondyloarthritis using the Spondyloarthritis research consortium of Canada scoring system. *BMC Musculoskelet Disord* 2022;23(1):654.
- [82] Wetterslev M, Østergaard M, Sørensen LJ, Weber U, Loft AG, Kollerup G, et al. Development and validation of 3 preliminary MRI sacroiliac joint composite structural damage scores in a 5-year longitudinal axial spondyloarthritis study. *J Rheumatol* 2021;48(10):1537–46.
- [83] Merkel PA, Herlyn K, Mahr AD, Neogi T, Seo P, Walsh M, et al. Progress towards a core set of outcome measures in small-vessel vasculitis. Report from OMERACT 9. *J Rheumatol* 2009;36(10):2362–8.
- [84] Exley AR, Bacon PA, Luqmani RA, Kitas GD, Gordon C, Savage CO, et al. Development and initial validation of the Vasculitis Damage Index for the standardized clinical assessment of damage in the systemic vasculitides. *Arthritis Rheum* 1997;40(2):371–80.
- [85] Seo P, Luqmani RA, Flossmann O, Hellmich B, Herlyn K, Hoffman GS, et al. The future of damage assessment in vasculitis. *J Rheumatol* 2007;34:1357–71.
- [86] Seo P, Merkel PA, Specks U, Hoffman GS, Langford CA, Spiera R, et al. Damage in ANCA-associated vasculitis: Preliminary evidence for the ANCA-associated vasculitis index of damage (AVID). *Arthritis Rheum* 2006;54(Suppl):S487.
- [87] Nakagomi D, Cousins C, Sznajd J, Furuta S, Mohammad AJ, Luqmani R, et al. Development of a score for assessment of radiologic damage in large-vessel vasculitis (combined arteritis damage score, CARDS). *Clin Exp Rheumatol* 2017; 35S103(1):139–45.
- [88] Seo P, Jayne D, Luqmani R, Merkel PA. Assessment of damage in vasculitis: expert ratings of damage. *Rheumatology (Oxford)* 2009;48(7):823–7.
- [89] Carruthers D, Bacon P. Activity, damage and outcome in systemic vasculitis. *Best Pract Res Clin Rheumatol* 2001;15(2):225–38.
- [90] Piga M, Floris A, Espinosa G, Serpa Pinto L, Koukias N, Lo Monaco A, et al. Development and preliminary validation of the Behçet's syndrome overall damage index (BODI). *RMD Open* 2020;6(2):e001192.
- [91] Gheita TA, Hammam N, Fawzy SM, Abd El-Latif E, El-Gazzar II. Egyptian College of Rheumatology-Behçet's Disease Study Group (ECR-BDSG). Development and validation of a Behçet's disease damage index for adults with BD: An explicit, composite and rated (ECR) tool. *Int J Rheum Dis* 2021;24(8):1071–9.
- [92] Gheita TA, El-Latif EA, El-Gazzar II, Samy N, Hammam N. Egyptian College of Rheumatology-Behçet's Disease Study Group (ECR-BDSG). Behçet's disease in Egypt: a multicenter nationwide study on 1526 adult patients and review of the literature. *Clin Rheumatol* 2019;38(9):2565–75.
- [93] Rabie M, Hassan R, Abdelhady EI, Kamal D. Damage accrual in Behçet disease: Behçet's syndrome overall damage index (BODI) versus Behçet's disease damage index (BDI). *Int J Rheum Dis* 2023; epub. ahead of print.
- [94] Gheita T, Hammam N. Machine Learning in Rheumatology: The emerging cutting-edge strategy. *Int J Clin Rheum* 2023;18(4):72–4.
- [95] de Punder YM, van Riel PL, Fransen J. A simplified baseline prediction model for joint damage progression in rheumatoid arthritis: a step toward personalized medicine. *J Rheumatol* 2015;42(3):391–7.
- [96] Ceccarelli F, Sciandrone M, Perricone C, Galvan G, Morelli F, Vicente LN, et al. Prediction of chronic damage in systemic lupus erythematosus by using machine-learning models. *PLoS One* 2017;12(3):e0174200.
- [97] Pego-Reigosa JM, Lois-Iglesias A, Rúa-Figueroa Í, Galindo M, Calvo-Alén J, de Uña-Álvarez J, et al. Relationship between damage clustering and mortality in systemic lupus erythematosus in early and late stages of the disease: cluster analyses in a large cohort from the Spanish Society of Rheumatology Lupus Registry. *Rheumatology (Oxford)* 2016;55(7):1243–50.
- [98] Kallas R, Li J, Goldman DW, Magder LS, Petri M. Trajectory of damage accrual in systemic lupus erythematosus based on ethnicity and socioeconomic factors. *J Rheumatol* 2022;49(11):1229–35.
- [99] To CH, Petri M. Is antibody clustering predictive of clinical subsets and damage in systemic lupus erythematosus? *Arthritis Rheum* 2005;52(12):4003–10.
- [100] Hammam N, Bakhiet A, El-Latif EA, El-Gazzar II, Samy N, Noor RAA, et al. Development of machine learning models for detection of vision threatening Behçet's disease (BD) using Egyptian college of rheumatology (ECR)-BD cohort. *BMC Med Inform Decis Mak* 2023;23(1):37.